

Lecture 18

Cybersecurity and Privacy - Extended Discussion

Cyber Security and Privacy | NPTEL 106106248

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Video: <https://www.youtube.com/watch?v=kZtw4L6LZS8>

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thank you for joining today's live session welcome and I'm hoping that all of you are enjoying the course and from the feedback that I get through our chat forum I noticed that most of the questions are related to platform rules as to what happens if I don't submit a week one assignment what happens if I miss two weeks assignment how is the evaluation going to be Etc which are already answers answered questions and I'm sure our team would also respond to such questions which are mostly generated to the related to the npt tunnel program rules okay I will not be specifically addressing them but I only address course related questions content mostly content course content related questions okay so one category of questions that is still repeating is about how the course material would be supplied and it appears that those who ask these questions have not gone through the live sessions okay particularly live session one addressed this question already okay and I will not repeat the same answers again and again in every live session okay so I request again those who have questions about reading materials please refer to live session one recorded video is already available in nptel please and let me also tell you that this is a postgraduate level course PG level course okay this is not an undergraduate level course okay typically in a ug level course A Faculty would use a textbook and explain the concepts in the textbook with say more examples and more elaboration in a classroom okay and that is how typically a undergraduate course will go in PG level and in PhD level okay we do not teach simply the textbook okay reading the textbook is the students or the participants responsibility and a class is not meant to repeat the same things which are already covered in the textbook an instructor May refer to the textbook but the class will be much more than the textbook okay because it is advanced level course and it is the same principle that is followed in the textbook in this course and that is why you see more reading materials that is used in the course because it's not a ug course it's a PG course and therefore I have brought course materials from other sources like journals case Publications by Harvard Business School and so on and therefore it is mandatory for this course to cover such current and advanced use such Advanced Materials to drive the concepts with touch with the world of practice okay and that is the reason it is the course uses those materials now your concern is how you get access to those materials for which I have given certain tips in live interaction one okay and please go through that and so and nptel policy will not allow supply of materials and textbooks for the participants you have to procure them yourself okay so I'm only giving on the rationale for why those materials are used in the course okay and yeah I I would leave that topic there let me also look at other questions that has come from the participants okay somebody has asked what is the full form of Bia I think that relates to week four contingency planning business impact analysis via okay and please read the textbook okay and that is very important for you to follow this course okay and the specific

chapters which I already I always cover or share with you in the live interaction which is the reading material that you should specifically go through and for week five I am going to share that with you today before I forget so give me a moment week five is about policy so let me actually share with you the specific reading from the textbook and other sources that you need to read in order to get a full view of the week okay so week five is about policy so here in the chat I am sharing the specific reading for that okay okay okay this is of course chapter four of your textbook has a section okay page number 177-213 covers information security policy okay then the lecture and the Powerpoints which is already shared with you covers more aspects of what is policy and what are the different levels in policy okay policy can be at a higher level policy can also operate at a procedural level okay so I would say what is changing is the level of abstraction okay at a higher level you you if the policy is for an organization then you connect the policy statements to the priorities and strategy of the organization okay and talk at that okay but when at a procedural level it comes to implementation how the specific aspects of the policy will be embedded say in certain rules say it could actually work at the level of a firewall configuration okay but it actually percolates from the top okay so there is a policy it gets implemented at a rule level so that is a basic idea which is explained in the course video and also the the Powerpoints will be useful please read the textbook chapter four Pages 177 to 213 and also you will you are given an additional reading which is from a article SBR article internet in security the end of cyber security okay that is another reading okay that is given for this course and please get access this material find a way to access this material on your own and I have given you tips in live interaction one okay that is as far as policy is concerned let me also check what are the other questions that has come from audience well I hope I am Audible and disabled everything is going well yes sir okay thank you okay there was a question oh okay there was a couple of errors that crept into the evaluation of week three or the assignment of week three which would be re-evaluated thank you for highlighting that I think many people wrote about this and we will be doing the re-evaluation okay sorry about it and and somebody has commented the lectures in each week enter abruptly and start at a different topic request continuity and proper closer in each lecture if possible since the participant has not given a specific example I am not able to understand what is the issue here for example each week has a topic okay and I am very sure about it each week has a topic for example week four the topic was contingency planning incidence response disaster recovery and business continuity okay so essentially it was about planning okay and one planning is contingency planning we will in the week six we will come to risk management which is planning to manage risk that is another type of plan okay between contingency planning and risk management you have the policy because it is policy that guides okay risk management so therefore we deal with that okay so therefore there is a rationale for the flow of the course each week okay so contingency planning was the theme of week four and there are three videos which shows the contingency planning topic is divided into three sub themes or subtopics and the final one would be a case discussion okay and you can feel you can feel or you can see that you know the case discussion has come abruptly okay it is not object okay it is a planned topic of the the case of the I Premier company which is a classical Harvard case denial of service attack which shows how an organization was not prepared okay to face an incident okay and it doesn't know how to respond to an incident

contingency planning is all about that okay you assume that despite all the protection mechanisms you have in place still things can go wrong okay and therefore what happens if you are not prepared okay you stare at the world okay you blame each other okay and you just see the clock ticking and no don't know what to do so that is the drama that you see a real drama that you see in the case Okay and then you ask this question how what the organization could have done or should have done to avoid this kind of issues what they should have done before what they should have done during and what they should have done post the incident okay and that is what contingency planning is basically the case is only driving the concepts of contingency planning in a with a practice point of view or with a real life point of view okay so each of these lectures are related look at the theme of the week and then try to see how the pieces connect well together okay and for example cyber security policy is the theme for week five and then you will see what is policy how policy changes Behavior at the even at the national level at the organizational level at the individual level and what are the different elements so basically it is to motivate the importance of policy that I first spend a few slides on you know how what is the importance of policy in the absence of policy you know the desired Behavior will not evolve okay if you have a desired behavior for an organization or for an individual okay and it is a policy that and the implementation of that policy that bring about a desired Behavior okay if security is an important compliance Behavior you accept from individuals then there should be a policy in place and the policy should be communicated and the policy should be implemented otherwise people will not comply okay so so that is the effort in this session along with a lot of examples from real life okay and then it is also driven by a article on internet insecurity okay the opposite of security so and how that article emphasizes the need for a informed policy okay especially when you have critical infrastructure and there is a specific meaning to critical infrastructure okay you have an industry talk coming up two weeks from now and then you will actually listen to what is critical infrastructure you know for example health health infrastructure okay or you your energy okay or Finance finance and banking Etc these are all critical infrastructure for a country okay and within within an organization you can identify critical infrastructure and that identification is important because unless you know what is critical to running your organization or to run your country you will not prioritize okay and when you prioritize well ABC are critical then have you actually plan for your security in the era of Internet okay and you will see that there is a very important observation by the author of that article that any system that is connected to Internet is insecure if you go by that observation then what do you do okay what do you connect and what you don't come when you decide to disconnect then there is an impact because you lose certain you you lose certain output or you lose certain efficiency you lose certain convenience of running you know your organization okay you seem to be turning the clock back and going back to the era of no internet is that the way so there is enough material for thought process there you know at a decision making level okay how do you prioritize and the subsequent we okay when it comes to risk management that is a week six you will see there is a real life case of a Manufacturing Company implementing iot solution and getting into problem in in into cyber insecurity okay so these are all sequence such that you understand contingencies you understand the importance of policy and subsequently you go into risk management okay so that is the topic for the next week okay so the current week is

on policy and there is a question on data owners on data custodians which I already clarified in the previous week and please do listen to the videos which are already there before firing your questions I'm sorry if you have asked these questions beforehand and somebody has asked how to control the smart cars I do not know without elaboration what do you mean by this question because smart cars are as you know Internet connected cars and therefore as I said any system that is connected to internet is not fully secure and therefore you are asking me a question as to how to solve this problem this is there is no you know there is no Panacea or you know a medicine sort of an advice sort of a thing to solve this problem one is you know the conventional or the conservative answer don't use Smart cards okay okay or don't use scars at all if you are afraid of accidents but then how can you use Smart cards and still be secure okay if that is your question then you may have to wait for week six where we address a similar problem which is related to iot Okay Internet connected sensors and controllers which are used in the industry how that can actually lead to problems or that that can lead to Cyber attack and if so what to do okay so the problems are similar so along with that case discussion you may get to know the different threats of the problem and what potentially could be a broad approach okay and I think I don't have any more I don't have any more questions that has come in the chat box and let me try to read out some questions which has just come up in the chat box now since I have exhausted the questions already which path one should follow to advance their career in infosec through certification courses like CH I think it stands for certificate on ethical hacking oscp and CI sssp it confuses me a lot while going through the dedicated path please help me okay so now so as I said in the beginning this is educational program or it's a course on cyber security and privacy it's a course it's not a it's not a training program which actually is intended for providing you certain skills okay this is not a skill based course it is actually knowledge okay and essentially a course an academy course would focus on educating you with Concepts okay and with understanding of those Concepts and Frameworks and how you can apply that okay now if you want to get trained in a particular specific area okay it could be training on say ISO standards or nist standards which is more management related it could be specifically on ceh oscp which I understand as technical skills you it it depends basically on your interest and what kind of job and career are you and looking for okay and since I myself having gone through a specific training of the kind which you mentioned here I haven't gone through a ethical hacking program myself I am my understanding is not at the awareness level I'm aware what it is but the specific benefits of it I'm not able to tell you of 100 okay so but always the best thing is that you connect with people who have already undergone such training programs and ask them what benefits they got if there is a community of trained people and the best way to connect with such people is platforms like LinkedIn okay so where you see a lot of promotions of this kind of programs and you will also see people have who have undergone this program so one suggestion is to have a LinkedIn account and then try connect with people who have gone through this training programs and what kind of careers they are in Etc okay to some extent you may get some insights when you have the industry talk two weeks from now okay so you know but not very specific to the particular training programs you have asked me okay and I do not see any Edge okay good so you know LinkedIn is the professional networking so it's through networking that you get to know a lot of things and then of course depending on which is the training Pro

which is the training organization right is it a circle or is it PMI okay isaka is very specifically focused on it's a global body I know it has a presence in India and they conduct training and their trainings are very valuable okay soccer trained would add definitely to your you know to your competence and also to your CV and you it I would recommend relevant training because depending on your interest and your career interest from isaka definitely since it is the most important industry body I would think for cyber security okay okay PMI would be another body and then of course the standard organizations like the iso which ISO doesn't directly give training but there are other organizations which provide ISO training okay so ISO a series of standards for cyber security we discussed already super I think if there are no further questions I can close here enjoy this week's videos and get get a firm footing on what is policy okay and what are different levels of policy what are the different levels at which a policy works this will be very important learning for you for not only for cyber security but to be a good professional you should be aware of the first document that you should ask for is what is a policy okay before making any subjective opinions all right so I will close here and then wish you all the best and see you next week foreign foreign there hello foreign